HSRW DATATHON 2026, KLEVE

From messy factory data to actionable insight

24 hours, 20 raw files and real production data from BBMED, a cosmetics manufacturer. What we found, and what we would tell the plant.

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THE CHALLENGE

One plant, three systems that don't agree

5

packaging lines: 2 bottle, 3 tube

20

raw files from MES, ERP, Excel and energy meters

43,848

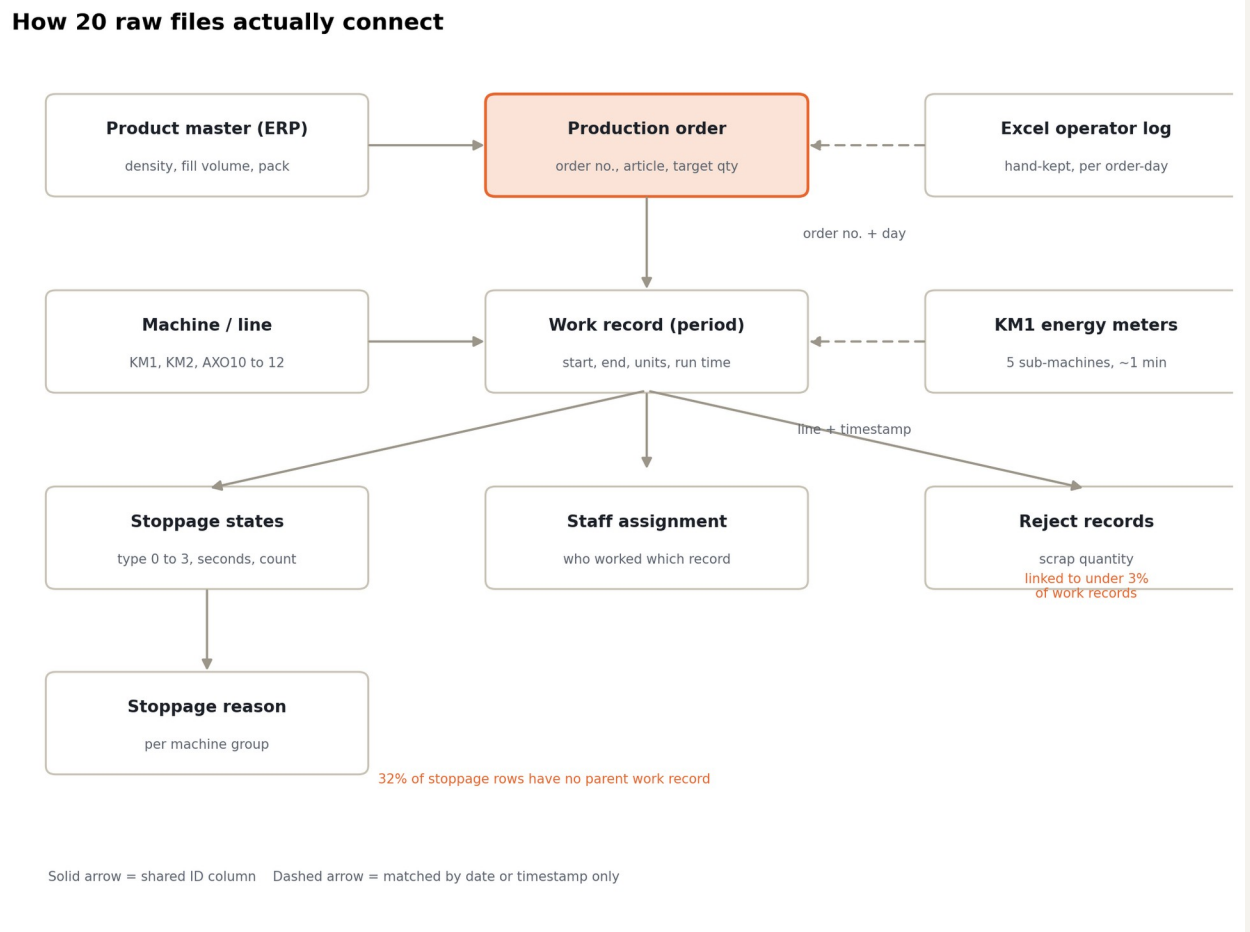
stoppage records, codes half-documented

24 h

from data release to jury presentation

The brief: understand performance, stoppages, staffing and energy on the lines, and fix how the data is collected. Every team got the same data. Saying this can't be answered yet was allowed, if we explained why.

How 20 raw files actually connect



STEP ONE

Before any chart: map how the tables connect

Joined by shared IDs

- Production order to work records (one per continuous run)
- Work records to stoppages, staff, rejects
- Article no. to ERP product master (density, volume)

Joined by time or date only

- Excel operator log: order no. plus calendar day
- KM1 energy meters: line plus timestamp
- No shared ID, so every match is a deliberate rule

Gaps we documented

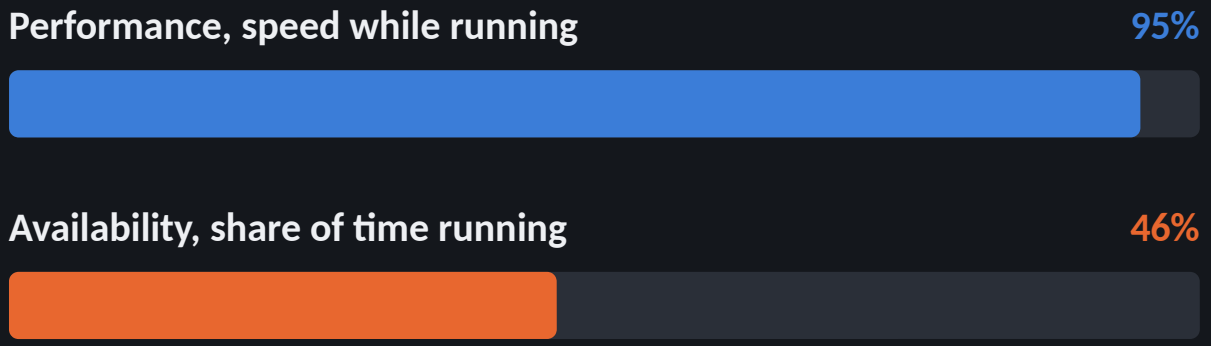
- 32% of stoppage rows have no parent record
- Scrap data on under 3% of work records
- 20 orders with stale target rates

FINDING 1, OEE

The machines are fast. They just don't run enough.

43.5%

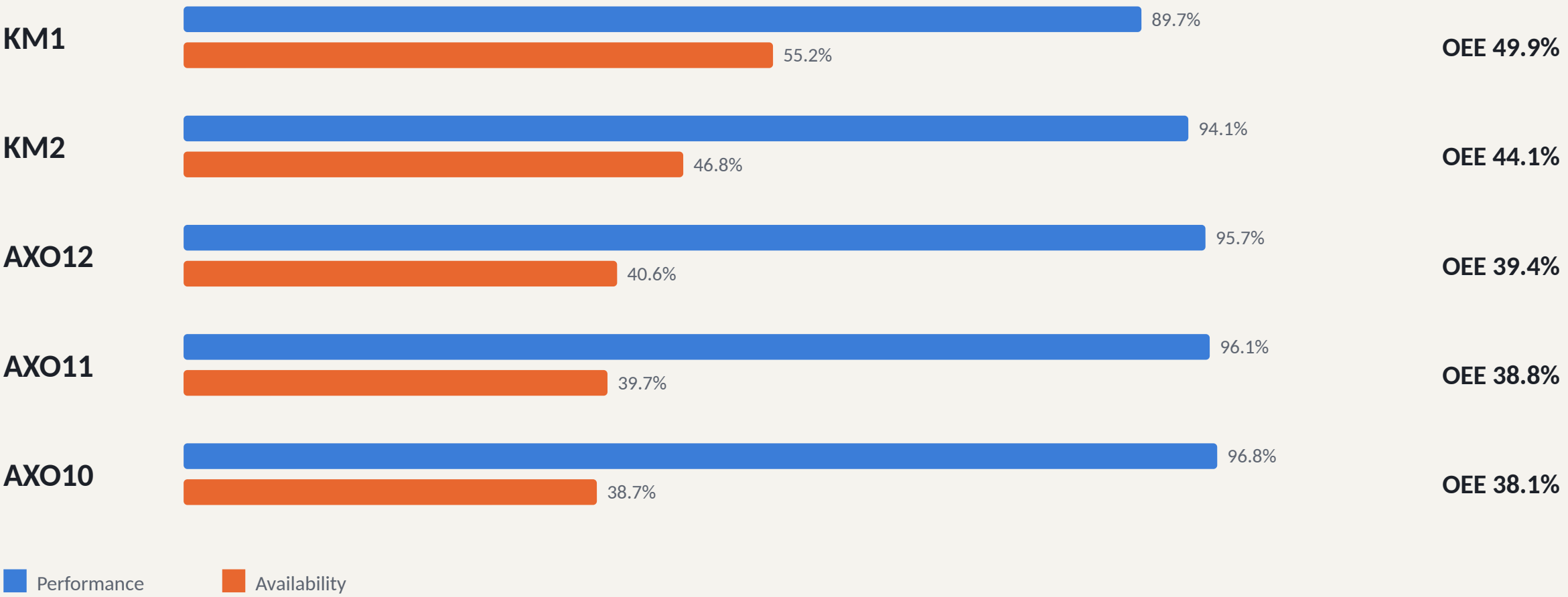
fleet OEE (availability times performance) across 889 clean orders



Every line hits 90 to 97% of target speed. The loss is time stopped, not slow running.

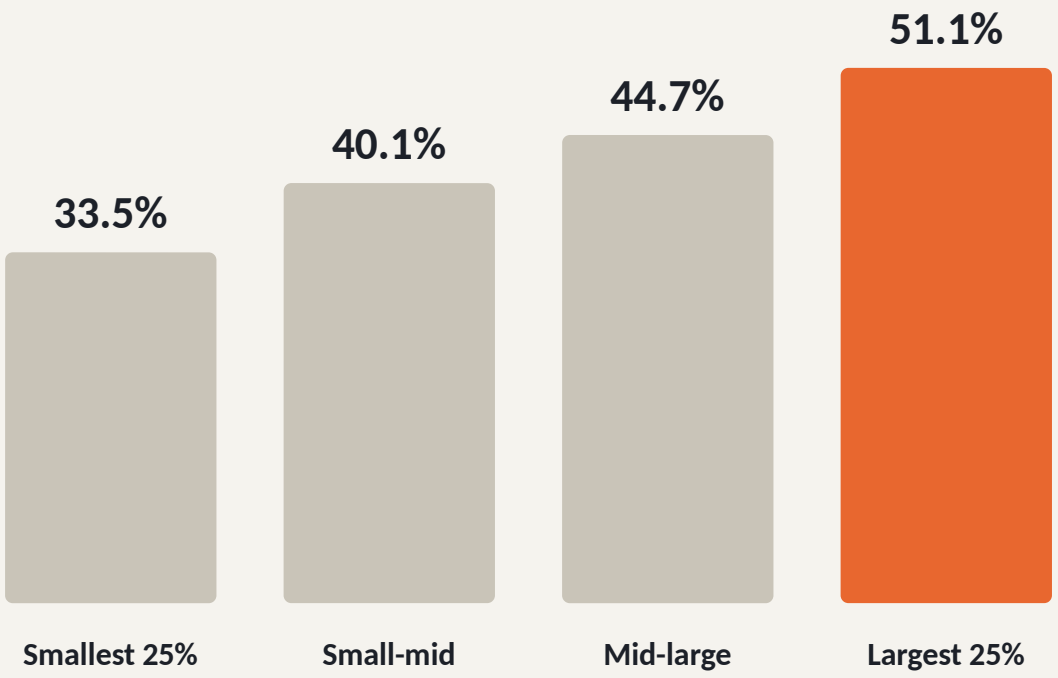
FINDING 1, BY LINE

Uptime decides the ranking, not speed



FINDING 2, ORDER SIZE

Bigger orders run better, purely through uptime



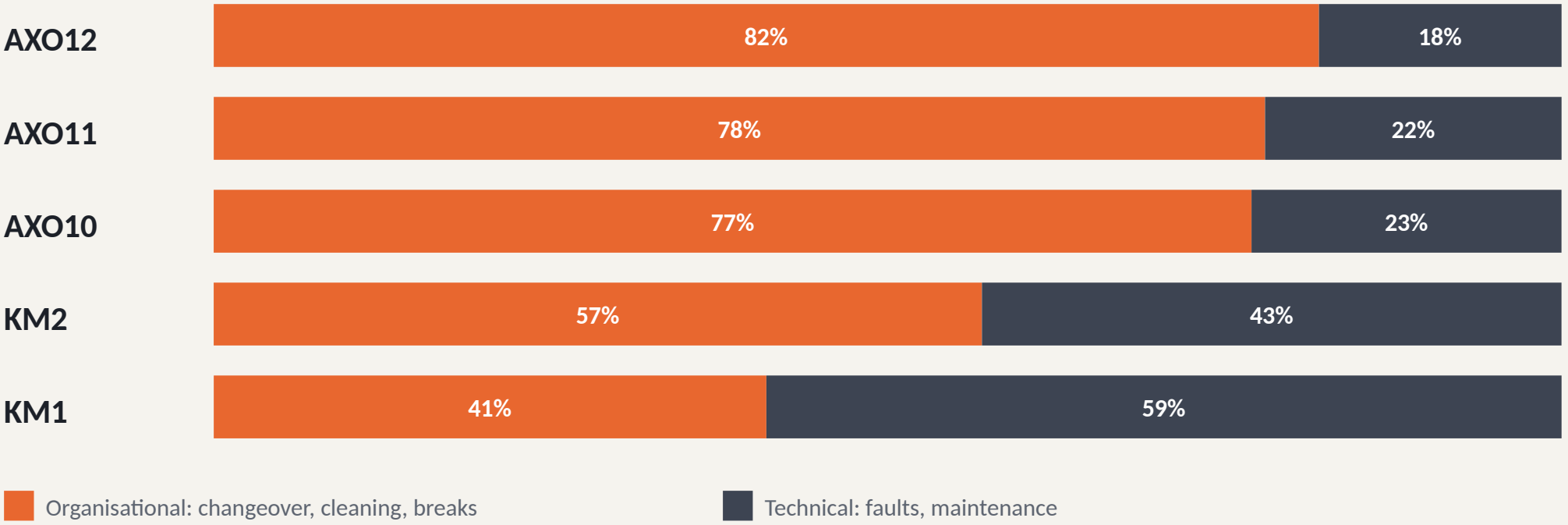
+17.6 pp

OEE gap between the smallest and largest quarter of orders. Speed is flat; fixed setup and cleaning time gets spread over more units.

Lever: batch small repeat orders.

FINDING 3, DOWNTIME

Lines stop for changeovers, not breakdowns



Setup and cleaning is the top stoppage reason on every line. KM1 is the exception, where labeller and carton-folder faults lead.

FINDING 4, DATA TRUST

Can the plant trust its own records? Mostly.

0.65%

median quantity gap between MES and the hand-kept Excel log. 90% of 263 shared orders agree within +/-10%.

9 orders drag the mean up to 30%

MES shows up to 55x the Excel quantity. Looks like record splitting, not a broken system.

Excel starts about 5 minutes earlier

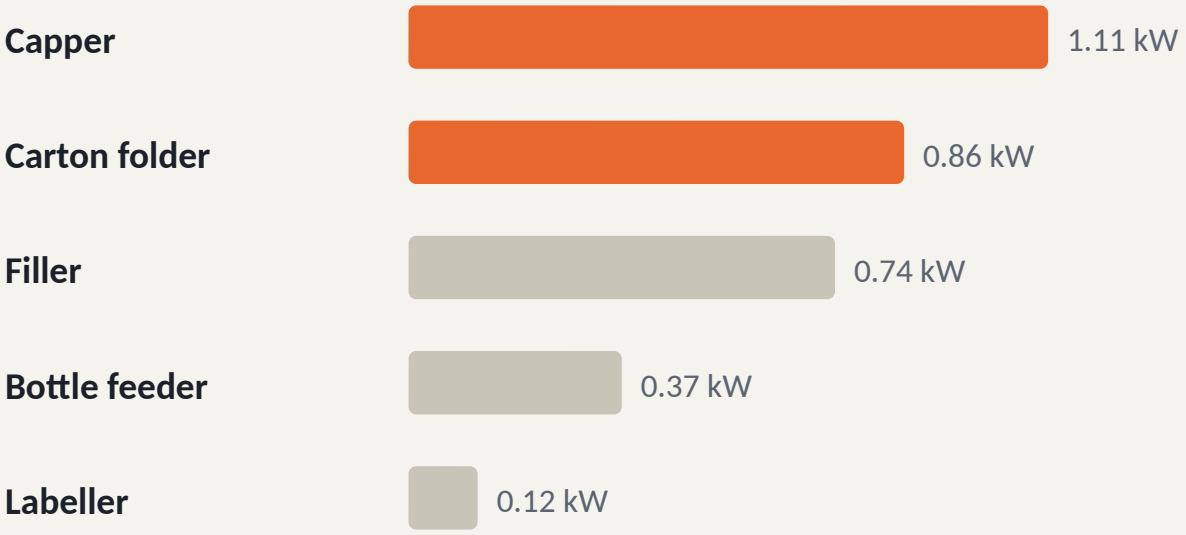
Operators write the shift start; MES logs the first booked minute.

MES keeps the clock running overnight

Durations run to 06:00 next day, inflating time and deflating rates.

FINDING 5, ENERGY ON KM1

Two sub-machines draw 61% of the line's power



5 meters, about 1 reading per minute, about 3.2 kW combined

ONE ORDER, TRACED END TO END

2.8 kWh

per 1,000 bottles, order 702026, 9,112 units

About a third of it was drawn while the line was stopped. Power barely moves (+/-5%) whether bottles flow or not.

FOLLOW-UP

The "broken" stoppage table, decoded

On the day, undefined codes and 18-day stoppages made the table look unusable. Tested record by record afterwards, it is a clean state-time ledger:

TYPE 3

Producing

Equals production time on 6,023 of 6,023 records

TYPE 1

Classified stops

One row per reason code, per machine group

TYPE 0

Micro-stops

Short stops never given a reason

TYPE 2

Idle before start

Matches the gap since the last run (r = 0.92)

RECOMMENDATIONS

What we would tell the plant

Batch small orders

The cheapest lever: +16 pp availability between smallest and largest orders.

Changeover study on the tube lines

77 to 82% of AXO downtime is organisational: setup, cleaning, staging.

KM1: standby and targeted maintenance

Idle standby for capper and folder; focus on labeller and folder faults.

Fix the data at the source

Order no. on every table, continuous energy logging, validation rules.

TAKEAWAYS

01 Map the data before you measure it.

02 Medians before means on operational data.

03 "The data is broken" is a hypothesis, not a conclusion.

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HSRW Datathon 2026, Kleve. Data provided by BBMED

Full case study and repo linked on LinkedIn